ARTIFICIAL INTELLIGENCE

CBSE Previous Year Question Paper 2024 • Q.P. Code 104 • Set 4
Complete Answer Key: MCQs, Short Answers & Long Answers

How to use this key: The paper asks candidates to attempt only selected questions. For revision, this document provides answers to all visible questions. Short and long answers are model answers; equivalent accurate wording is acceptable.

SECTION A — OBJECTIVE TYPE QUESTIONS (MCQs)

Q1. Choose the correct option

1(i) (c) Achievable — the ‘A’ in SMART commonly stands for Achievable.

1(ii) (b) Not taking feedback — self-management includes planning, setting goals and adapting; refusing feedback is not a key element.

1(iii) (a) Hard working — perseverance and effort are qualities of successful entrepreneurs.

1(iv) (b) Operating System — it manages computer hardware and provides services for applications.

1(v) (b) Public communication — one person addressing a large gathering is public communication.

1(vi) (a) Both A and R are true, and R correctly explains A — organic farming is a green skill because it promotes environmentally friendly, sustainable practices.

Q2. Choose the correct option

2(i) (b) Relevant and Authentic Training Data — useful, representative data supports a reliable AI project.

2(ii) (d) Automatic Summarisation — it condenses information from multiple sources while reducing repetition.

2(iii) (d) A digital alarm clock that rings at a set time every morning — it follows a fixed rule and does not need AI.

2(iv) (a) Care — lemmatisation converts ‘caring’ to its base dictionary form, ‘care’.

2(v) (c) Self-awareness, from recognising weaknesses and strengths to recognising one’s own feelings — this describes intrapersonal intelligence.

2(vi) (c) SQL — SQL is a database query language, not a tabular data-file format like CSV or a spreadsheet.

Q3. Choose the correct option

3(i) (a) Precision — the fraction of predicted positive cases that are actually positive.

3(ii) (d) Data Acquisition → Data Exploration → Problem Scoping → Modelling → Evaluation is NOT the standard sequence. The correct AI project cycle is Problem Scoping → Data Acquisition → Data Exploration → Modelling → Evaluation. Therefore, among the printed options, none appears to give the correct sequence exactly; option (a) begins with Data Exploration and is also incorrect.

3(iii) (c) Data Science — it combines statistics, data analysis and machine learning to understand real-world phenomena.

3(iv) (c) Instance Segmentation — it identifies individual objects and labels the pixels belonging to each instance.

3(v) (b) When a spam email is mistakenly identified as legitimate — the model fails to detect spam, producing a false negative.

3(vi) (d) Stock Market Analysis — it is not typically classified as an NLP application; the other options directly involve language or speech.

Q4. Choose the correct option

4(i) (a) Both Statement 1 and Statement 2 are correct — a confusion matrix is a table used to evaluate classification performance.

4(ii) (a) Clustering — the diagram shows two groups of data points formed without labelled categories.

4(iii) (c) Natural Language Processing (NLP) — Bag of Words is a text representation method used in NLP.

4(iv) (d) A music streaming platform that suggests songs and playlists based on listening history — this is a recommendation system.

4(v) Any two: Google and Bing (other valid examples include Yahoo and DuckDuckGo).

4(vi) (c) To assess how well the chosen model will work in future — evaluation estimates how well a trained model performs on suitable test data.

Q5. Choose the correct option

5(i) (a) Both A and R are true, and R correctly explains A — image resolution is related to pixel count, commonly calculated as width × height.

5(ii) (c) Artificial Intelligence — the ability to learn, make decisions and improve in ways that mimic human intelligence is described as AI.

5(iii) (c) Statement 1 is correct, but Statement 2 is incorrect — precision or recall can be used to evaluate performance, while if both precision and recall equal 1, the F1 score is 1, not 0.

5(iv) (b) Computer Vision — face filters detect and track facial features in images or video.

5(v) (a) When — the standard 4Ws Problem Canvas uses Who, What, Where and Why; 'When' is not one of those four blocks.

5(vi) (c) Natural Language Processing (NLP) — virtual assistants such as Siri use NLP to interpret and respond to spoken or written requests.

SECTION B — SUBJECTIVE TYPE QUESTIONS

Short Answers

Q6–Q10: Employability Skills (2 marks each)

Q6. Give any two examples of how individual choices and behaviours can contribute to sustainable development.

Use reusable bags and bottles instead of single-use plastic; conserve electricity and water by switching off unused lights and closing taps.

Q7. List any two common misconceptions about entrepreneurship.

1. Entrepreneurs are born with all the required skills and cannot learn them. 2. Entrepreneurs must start with a large amount of money. In reality, skills can be developed and many businesses begin small.

Q8. What is the importance of time management in effectively dealing with stress? Provide one strategy.

Time management reduces last-minute pressure by helping people prioritise and complete tasks on time. One strategy is to make a daily to-do list and rank tasks by urgency and importance.

Q9. Mention any two measures to protect data from theft and viruses.

Use updated antivirus/security software and keep the operating system updated. Also use strong, unique passwords and avoid suspicious links or attachments; regular backups provide additional protection.

Q10. How can communication choices affect peer, family and customer relationships? Give four factors for choosing the right method.

The method affects clarity, tone, speed and the chance of misunderstanding. Choose based on: (1) purpose of the message, (2) audience and relationship, (3) urgency, and (4) complexity/confidentiality of the information.

Q11–Q16: Artificial Intelligence (2 marks each; answer in about 20–30 words)

Q11. Differentiate between Machine Learning (ML) and Deep Learning (DL).

Machine Learning enables systems to learn patterns from data using algorithms. Deep Learning is a subset of ML that uses multi-layer neural networks and often needs more data and computing power.

Q12. What are the primary differences between Script-bots and Smart-bots?

Script-bots follow predefined rules and fixed scripts. Smart-bots use AI techniques such as NLP or machine learning to interpret user input and respond more flexibly.

Q13. What do you mean by Evaluation of an AI Model? Explain overfitting.

Model evaluation measures performance on unseen test data using metrics such as accuracy, precision and recall. Overfitting occurs when a model learns training data too closely and performs poorly on new data.

Q14. Why would a healthcare organisation use unsupervised learning to predict disease outbreaks from medical records? Briefly explain.

Unsupervised learning can find hidden patterns, clusters and unusual trends in records when labelled outbreak outcomes are unavailable. These patterns can help flag possible emerging risks for expert review.

Q15. What role does data play in AI applications? Name two online data-collection sources.

Data is the material from which AI systems learn patterns, make predictions and improve. Online sources include web forms/surveys and public datasets or website/API data.

Q16. Differentiate between grayscale and RGB images.

A grayscale image represents each pixel with one intensity value, from dark to light. An RGB image uses three channel values—red, green and blue—to represent colour.

LONG ANSWERS (50–80 WORDS EACH)

Q17. What are Neural Networks? Briefly explain all the layers of a neural network.

A neural network is an AI model inspired by how connected neurons process information. The **input layer** receives the data or features. One or more **hidden layers** process the inputs using weighted connections, biases and activation functions to learn patterns. The **output layer** produces the result, such as a predicted class or numerical value. During training, weights are adjusted to reduce prediction error.

Q18. Give four examples of applications of AI that we see around us.

Four everyday applications of AI are: (1) virtual assistants such as Siri or Google Assistant, which interpret requests; (2) recommendation systems on music and video platforms; (3) navigation apps that estimate traffic and suggest routes; and (4) email spam filters that classify unwanted messages. AI is also used in face recognition, translation, customer-support chatbots and medical-image analysis.

Q19. Documents: (1) "ML and DL are part of AI." (2) "DL is a subset of ML." Implement the Bag of Words model and depict the outcome of each step.

Step 1 — lowercase and remove punctuation. Document 1 becomes: "ml and dl are part of ai"; Document 2 becomes: "dl is a subset of ml". Step 2 — tokenize the words. Step 3 — create a vocabulary in first-occurrence order: **[ml, and, dl, are, part, of, ai, is, a, subset]**. Step 4 — count each vocabulary word in each document. The resulting vectors are shown in the table below.

Vocabulary word	Doc 1	Doc 2
ml	1	1
and	1	0
dl	1	1
are	1	0
part	1	0
of	1	0
ai	1	0
is	0	1
a	0	1
subset	0	1

Q20. Identify the two supervised-learning models in Figures 1 and 2 and give suitable examples.

Figure 1 shows **classification**: a decision boundary separates data points into classes. Example: classifying an email as spam or not spam. Figure 2 shows **regression**: a fitted curve models a continuous numerical value. Example: predicting a house price from its size. Classification predicts a category or label, whereas regression predicts a number.

Q21. Confusion matrix: calculate True Negatives, Precision, Recall and F1-score.

From the table, treating "Fake News = Yes" as the positive class: TP = 45, FP = 15, FN = 20 and TN = 420. Therefore, True Negatives = **420**. Precision = TP/(TP+FP) = 45/60 = **75%**. Recall = TP/(TP+FN) = 45/65 = **69.23%**. F1-score = 2PR/(P+R) = 90/125 = **72%**.

Important calculation notes

• Q21 confusion matrix is read with rows as Predicted (Yes/No) and columns as Reality (Yes/No), as printed on page 11. Thus TP=45, FP=15, FN=20, TN=420.
• For Q3(ii), the printed options appear to omit the standard AI project-cycle order. The standard order is Problem Scoping → Data Acquisition → Data Exploration → Modelling → Evaluation; verify with the prescribed textbook if the exam expects a particular option.

Source basis: answers are prepared from the uploaded 2024 question paper (11 pages), including the figures on pages 7, 10 and the confusion matrix on page 11. Where an answer depends on standard AI curriculum terminology, the response is phrased as a model answer for revision.